

● Section VI - Contact Diseases: Tetanus + Rabies		
Source / scope	L4 - NUT401 - Tetanus & Rabies Dr. Inas PDF. Original scanned textbook pages 214-222. All scientific facts, definitions, tables/flowcharts, regimens, numbers, doses, risk groups, and control/prevention points are preserved and reorganized.	
Visuals to recreate	Tetanus wound-management flowchart (immunity categories A-D + clean vs other wounds) and Rabies post-exposure prophylaxis table are rebuilt as compact text tables.	

● TETANUS - overview, agent, source, infection		
Definition / importance	Tetanus is an acute disease. Prevention has high public-health priority. It is not transmissible from one person to another. It has a high case-fatality ratio. It is almost completely preventable.	
Causative agent	<i>Clostridium tetani</i> (<i>C. tetani</i>): anaerobic gram-positive rod existing in vegetative and sporulated forms. Spores are highly resistant to heat and chemical agents but destroyed by sterilization procedures.	
Spores / toxin	Spores survive for years in suitable environmental conditions; found in soil, dust, animal faeces, less commonly human faeces, and on human skin. Germination/multiplication are favored by necrotic tissue and lack of oxygen at inoculation site. <i>C. tetani</i> produces exotoxin tetanospasmin , a potent selective neurotoxin responsible for clinical tetanus.	
Sources of infection	Organisms are present in the gastrointestinal tract of man and animals, e.g., horses, cattle and other herbivorous animals; they exit with faeces contaminating soil.	
Modes of infection	1) Contamination of wounds/abrasions with spores in soil. 2) Infected catgut (unsterile surgery). 3) Tetanus neonatorum from infection of the umbilical stump. 4) Puerperal tetani after abortion and childbirth.	

+ Tetanus - clinical picture, diagnosis, epidemiology		
Incubation	1-3 weeks .	
Symptoms/signs	Slight irritability at wound site. Powerful exotoxin causes strong tonic muscle contractions that may start in the masseters causing lock jaw and spread to the entire muscle system. In severe cases coma and death may occur.	
Diagnosis	Clinical picture is sufficient, or organisms may be isolated from wound exudate.	
Communicability	None. Disease is not transmitted from person to person.	
Epidemiology	May occur sporadically; more common in groups liable to infection: children, farmers, gardeners and soldiers. In some developing countries tetanus neonatorum remains a major problem. Higher incidence in males, but women are more exposed during delivery/abortion. No age is immune unless protected by previous immunization.	

■ Tetanus prevention - active immunization		
Principle	Best prevented by active immunization with tetanus toxoid. All persons should be immunized regardless of age.	
Preparations	1) Combined vaccine DPT. 2) Monovalent vaccine: plain toxoid or purified tetanus toxoid (adsorbed).	
Combined vaccine is given to	- Compulsory to infants at 2, 4, 6 and 18 months in the EPI. - At school entry: booster dose of DT. - After 10 years: third booster dose of TT vaccine. - Adults: monovalent TT given to soldiers and gardeners; primary course = two doses of TT adsorbed one month apart; one booster after one year. - TT also given to pregnant female to prevent tetanus neonatorum.	

■ Tetanus wound management - clean vs other wounds		
Immunity category	Clean, non-penetrating, negligible tissue damage / < 6 h old	Other wounds
A	Nothing more required	Nothing more required
B	Toxoid 1 dose	Toxoid 1 dose
C	Toxoid 1 dose	Toxoid 1 dose + Human TIG
D	Toxoid complete course	Toxoid complete course + TIG*
A	Complete toxoid course or booster within past 5 years.	
B	Complete toxoid course or booster >5 years ago and <10 years ago.	
C	Complete toxoid course or booster >10 years ago.	
D	Complete toxoid course OR immunity status unknown.	
* TIG	Tetanus Immunoglobulin.	

■ Tetanus control + neonatal tetanus		
Control - patients	- Notification to local health authorities. - Treatment by antitoxin in a single large dose up to 50000 I.U.; better given I.V. with penicillin. - Isolation in hospital in case of fever is preferred.	
Neonatal tetanus	Killer disease second to measles among six target diseases of EPI. Kills about 85% . Global mortality rate from tetanus neonatorum = 6.8/1000 live births . Common cause: infection of umbilical stump after birth. Starts 7 days after birth . Immunized mothers transmit antibodies to newborn protecting against disease.	
Maternal/infant protection	At least 2 doses : first dose at 16 weeks ; second dose at least 4 weeks after first dose and at least 3 weeks before delivery .	
WHO elimination principles	Neonatal tetanus targeted by WHO for elimination. Strategic principles: 1) High vaccine coverage of pregnant women. 2) Provision of hygienic clean delivery medium. 3) Health education of birth attendants. 4) Immunization of child-bearing women.	

● RABIES - overview, reservoir, transmission		
Definition	Rabies is a zoonotic disease transmitted from animals to man. It is an acute viral infection of the central nervous system (CNS).	
Causative organism	A rhabdovirus.	
Reservoir	Dogs, cats and wildlife animals: wolves, foxes and bats.	
Occurrence	50,000 human deaths/year worldwide .	
Epidemiologic pattern	1) Urban type in dogs and cats. 2) Wildlife rabies in wolves, foxes and bats.	
Modes of infection	Rabies virus may be present 7 days before symptoms in dogs. Transmitted by: 1) bites of infected dogs; 2) respiratory rabies (rare) in bat caves or laboratory work during experiments on rabies virus. Person-to-person transmission is theoretically possible because saliva of infected humans may contain virus, but has never been documented.	

+ Rabies - disease in animals and humans		
Disease in animals	Incubation period: 20 to 60 days or more . 1) Furious rabies: dog is easily irritated. 2) Dumb/paralytic rabies: dog tends to hide and gets paralyzed. Clinical picture depends on dose of virus and its strain.	
Disease in humans	Incubation varies and depends on severity of wound and site of bite, because virus travels along nerves to CNS. Incubation is short for bites of head and neck and long in extremities.	
Human incubation period	Varies from 10 days to 15 months , average 6 weeks .	
Clinical picture	1) Hydrophobia due to difficult swallowing and painful muscle contraction (muscles of deglutition); reflex contraction at sight of liquids. 2) Alternate periods of excitability up to mania and quietness. 3) Paralytic manifestations are late or absent.	
Diagnosis	History of exposure and clinical picture. If animal is healthy, keep under supervision for 10 days for rabies symptoms. If symptoms develop, animal should be killed and brain examined.	

■ Rabies prevention - post-exposure wound management		
Immediate wound care	Prompt and adequate treatment of all skin wounds possibly contaminated with rabies virus (animal bites/scratches). 1) Thorough cleansing with 20% soap solution or detergent and flushing with water, or cauterization with nitric acid (same effect as soap). 2) Wound should not be sutured immediately. 3) Antibiotics and tetanus prophylaxis may be indicated.	
Dog observation rule	If dog did not show clinical picture of rabies within 10 days , no post-exposure anti-rabies immunization is needed. If animal is rabid or escaped, bitten person should receive anti-rabies immunization.	

■ Rabies post-exposure immunization + vaccines		
Old vaccines	1) Semple vaccine: phenol preserved nerve tissue vaccine. 2) Suckling mouse vaccine. 3) Duck-embryo vaccine. Old types lead to hypersensitivity reactions.	
Recent vaccines	1) Human diploid cell vaccine (HDCV): gives rapid, high and greater conversion of antibody titre; less serious reactions and less common side effects: headache, nausea, vomiting, abdominal pain, muscle pain and dizziness. 2) Rabies vaccine adsorbed (RVA) used since the 1980s.	
Both vaccine regimens	5 doses , each 1 ml , intramuscular in deltoid region, or in infants in antero-lateral upper thigh; never in gluteal region.	
If animal is rabid	First dose as soon as possible after bite, then days 3, 7, 14 and 28 after first dose.	

■ Rabies immune globulin (RIG) + PEP recommendations		
RIG rules	Rabies immune globulin (RIG) = passive immunization; human origin. Give to all humans bitten by animals whose rabies cannot be excluded (e.g., animal escaped). Not indicated for those who received pre-exposure vaccine. Given ASAP up to the 8th day of starting vaccination regimen. Half dose infiltrated into wound and other half IM. Neutralizes virus in bite wound.	
Animal	Evaluation	Post-exposure recommendations
Dogs & cats	Healthy available	Should not begin prophylaxis unless animal symptoms develop.
Dogs & cats	Rabid or suspicious	Immediate vaccination + RIG + HDCV.
Dogs & cats	Unknown or animal escaped	Immediate vaccination and consult health office.
Wildlife animal	Any exposure / unknown	Immediate vaccination and consult health office.

◆ Rabies pre-exposure immunization + control		
Pre-exposure groups	High-risk groups: veterinarians, laboratory workers with rabies, dog handlers, field workers. HDCV or RVA vaccines should be given. No RIG is given in this regimen.	
If bitten after pre-exposure	Need post-exposure dose if bitten by rabid animal: 3 doses required on 0, 7, 21 or 28 day .	
Antibody titre	Periodically examined; if declines to less than 1:5 , booster dose should be given.	
Control of rabies	All stray dogs and cats should be killed. Pet dogs should be immunized and receive booster doses regularly. Health office should be informed about any rabid animal or bitten human. High-risk groups should receive pre-exposure immunization and boosters if antibody titre declines.	

★ Quick facts to memorize		
Tetanus	1-3 weeks incubation; DPT at 2,4,6,18 months; booster at school entry; 10-year booster; TT adult 2 doses one month apart + booster after one year; wound categories A-D; antitoxin up to 50000 I.U.; neonatal tetanus kills 85%; 6.8/1000 live births; starts 7 days after birth; maternal TT: first 16 weeks, second >=4 weeks later and >=3 weeks before delivery.	
Rabies	50,000 deaths/year; dog virus present 7 days before symptoms; animal incubation 20-60 days or more; human incubation 10 days-15 months, average 6 weeks; observe animal 10 days; 20% soap solution; vaccine days 0,3,7,14,28; RIG up to 8th day; pre-exposure post-bite doses 0,7,21 or 28; titre threshold <1:5.	